24. DECISION TREE USING SCIKIT-LEARN WITH VISUALISATION

Aim

To build and visualise a Decision Tree Classifier using the scikit-learn library on the Iris dataset and evaluate its classification accuracy.

Algorithm + Procedure

1. Import the required libraries from scikit-learn and matplotlib.
2. Load the Iris dataset using load_iris().
3. Separate the dataset into input features X and target classes y.
4. Split the dataset into training and testing sets using train_test_split().
5. Create a DecisionTreeClassifier using the entropy criterion.
6. Train the decision tree using the training dataset.
7. Predict the target classes for the test dataset.
8. Calculate the accuracy of the trained classifier.
9. Visualise the decision tree using plot_tree().
10. Display the decision tree and classification accuracy.

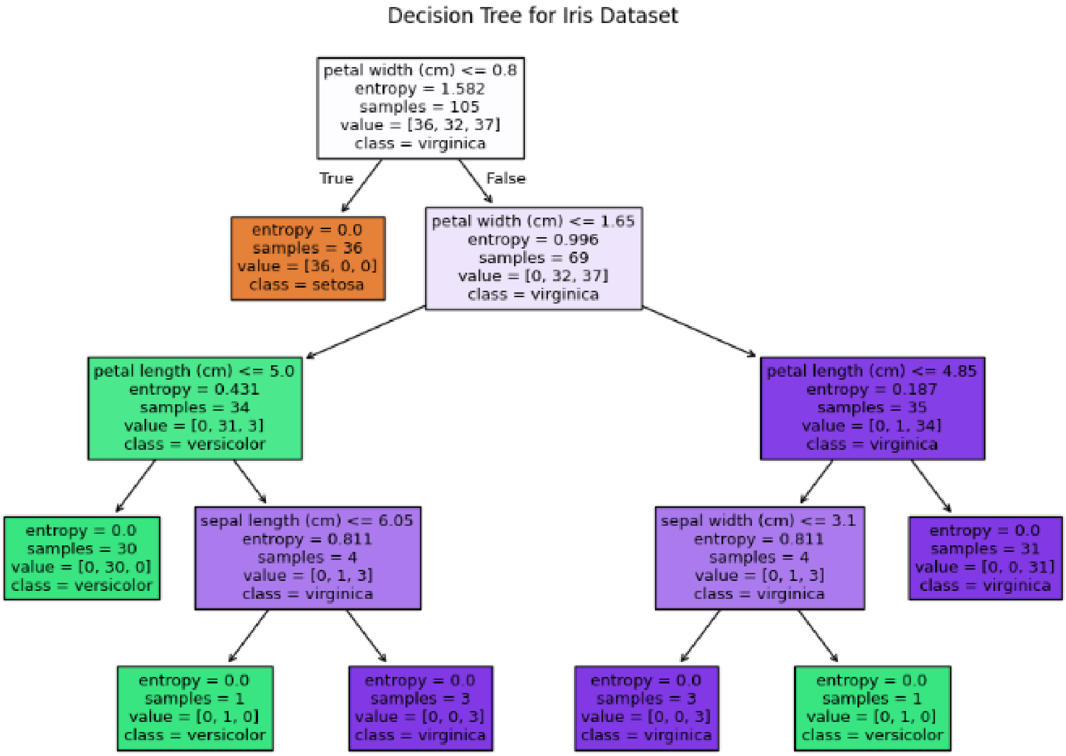
Code

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier, plot_tree

from sklearn.metrics import accuracy_score

import matplotlib.pyplot as plt

# Load Iris dataset

iris = load_iris()

X = iris.data

y = iris.target

# Split dataset into training and testing data

X_train, X_test, y_train, y_test = train_test_split(

    X,

    y,
```

```
    test_size=0.3,
    random_state=1
)
# Create Decision Tree Classifier
clf = DecisionTreeClassifier(
    criterion='entropy',
    random_state=1
)
# Train the model
clf.fit(X_train, y_train)
# Predict test data
y_pred = clf.predict(X_test)
# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
# Visualise the decision tree
plt.figure(figsize=(12, 8))
plot_tree(
    clf,
    feature_names=iris.feature_names,
    class_names=iris.target_names,
    filled=True
)
plt.title("Decision Tree for Iris Dataset")
plt.show()
```

Output:

```
Accuracy: 0.9555555555555556
```



Result:

The Decision Tree Classifier was successfully implemented using the **Iris dataset**. The model achieved an accuracy of approximately **95.56%**, and the trained decision tree was successfully visualised.